Lesson 1.13-1.14: Function Models and Applications

Name_____ Date _____ Period _____

Problems 1 – 4, In each data set below, state the type of model (linear, quadratic, cubic, piecewise) that best represents the data and give an explanation to justify why the model fits.

1.

x	1.2	3	4.8	6.6	8.4
$f(x)$	2.2	3.1	4	4.9	5.8

2.

x	-3	-2	-1	0	1
$f(x)$	-2	-5	-6	-5	-2

3.

x	-4	-2	0	2	4	6
$f(x)$	2	3	4	0	-4	-8

4.

x	0.8	1.5	3	4.8	7
$f(x)$	9.5	8.3	7.2	5.5	2.5

Problems 5 – 7, use a graphing calculator to find a regression model to predict.



5. The table at right, shows the amount of money, in billions of dollars, spent on public education from 2010 to 2014.

US Education Spending	
Year	$ in billions
2010	919.67
2011	903.31
2012	920.57
2013	878.23
2014	883.00

Source: www.usgovspending.com

A. Write the linear regression model for the data, where $t = 0$ represents 2010.

B. What is the average rate of change of spending from 2011 to 2012?

C. Use the equation from part (A) to estimate projected spending for 2018.

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6. A copier cost $7650 when purchased. After 3 years its value is $3825.



A. Write a linear model to represent the scenario.

B. Use your model to predict the value of the copier after 4 years.

C. What would be a reasonable domain for the model? Why?

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7. The table lists the cost of 2% milk. Use a scatter plot to:



A. Find the linear regression equation of the data.

B. Use the model to predict the price of milk in 2015.

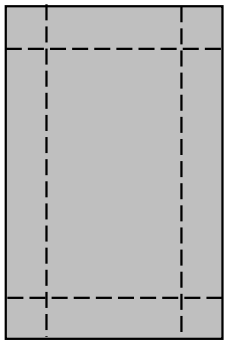
C. Based on your linear model, how much would you expect to pay for a gallon of 2% milk in 2020? Why might the estimate not be reasonable?

Cost of 2% Milk	
Cost of 2% Milk Year	Average Cost per Gallon
1998	$2.57
2000	$2.83
2002	$2.93
2004	$2.93
2006	$3.10
2008	$3.71
2012	$3.87

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8. A square of side x inches is cut out of each corner of a 12in. by 15in. piece of cardboard and the sides are folded up to form an open-topped box.

A. Write the volume V as a function of its side x.



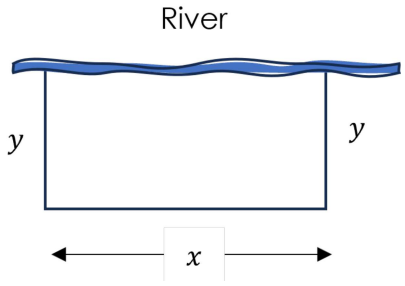
B. What is a reasonable domain for the model? Why?

C. Use a graphing utility to find the value of x that will maximize the volume.

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9. Ryan has 3000 feet of fencing to enclose a rectangular field for his horses. One side of the field lies along the river, so only three sides require fence material (see image below).

A. Express the area A of the rectangle as function of the width x of the rectangle.



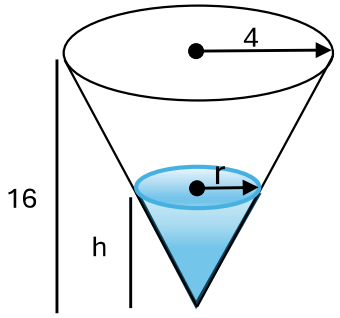
B. What is the domain of A?

C. Use a graphing utility to find the value of x that produces the largest area.

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10. Water is poured into a container in the shape of a right circular cone with radius 4 inches and height 16 inches. Express the volume V of the water in the cone as a function of the height h of the water.

$V = \frac{1}{3}\pi r^2 h$



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11. The table below gives the stopping distance for an automobile under certain road conditions.

Speed (mi/hr)	20	30	40	50	55
Stopping Distance (ft)	18	37	68	105	128

A. Find a linear model for the data.

B. Find a quadratic model for the data.

C. Compare the models. Which is better? Explain your answer using characteristics from the data in the table.

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12. Amy has been a realtor for 27 years. The number of sales for homes are recorded in the table below.

Year	1983	1985	1989	1991	1996	2003	2009	2010
Sales	15	18	21	25	28	31	35	37

A. Find a quadratic model for the data. Use 1983 as year 0.

B. Estimate the number of sales in the year 2000.

C. Estimate when there were 32 home sales.

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